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P2P Settlement — Wallet-Native Cross-Rail Swap

What this is. A peer-to-peer swap between crypto on **Bigtangle L0** and fiat on **PayPal** (or the CNY rails — WeChat Pay / Alipay / bank). The seller escrows tokens in a **2-of-3 P2SH address** (seller, buyer, engine); the buyer pays an exact-amount fiat obligation; the engine proves the payment and the escrow releases. Every irreversible step is signed with the wallet's own PQ key, and the engine can only ever **co-sign** a spend — it never holds your funds.

Where. The **P2P** screen in the wallet (sidebar → Trade → P2P). Everything below is captured from that screen on the local demo build.

Demo vs. live. The screenshots come from the demo build: the engine runs in mock-PayPal mode (stubbed invoice URLs and synthetic transaction hashes) and the lock/verify/release legs are signed with a local engine key. The state machine, the signatures, the append-only event log and the on-chain audit anchor are the real thing; only the external PayPal calls and the L0 broadcast are stubbed.

No manual "Confirm". There is no seller "confirm" button on the PayPal rail — the engine verifies the payment from its own evidence. (The CNY rails, which have no webhook, use an explicit seller confirmation instead; see `docs/p2pcny.md`.)

The flow at a glance

ACTIVE —match→ MATCHED —lock→ ESCROW_LOCKED —payment→ PAYMENT_PENDING

|

PAYMENT_PENDING —verify→ PAYMENT_VERIFIED —release→ ESCROW_RELEASED —payout→ COMPLETED

#	Step	Who acts	On-chain / engine
1	Seller lists a signed sell order	seller	order stored, not funded
2	Buyer matches (receive address + PayPal account)	buyer	swapId created, escrow address derived
3	Seller funds the 2-of-3 escrow, engine proves the lock	seller + engine	ESCROW_LOCKED on L0
4	Buyer pays the exact-amount invoice and reports it	buyer	PAYMENT_PENDING
5	Engine verifies the payment itself	engine	PAYMENT_VERIFIED
6	Engine + buyer co-sign the release, tokens move	engine + buyer	ESCROW_RELEASED
7	Engine pays the seller via PayPal Payouts	engine	COMPLETED

Every transition is anchored as a `social.p2p-swap` record on the L1-SOCIAL chain, so the whole lifecycle is publicly auditable without exposing either party's PayPal details.

1. The seller lists a sell order

On the **Open sells** tab, fill in the order: the token and amount you give, the fiat price you want, the currency, the chain the tokens sit on, and the ****payment method**** (PayPal or a CNY rail). Terms are locked into the order when you list it, so this is signed with your wallet key.

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Open sells

My swaps

Refresh

List a sell order

Token

USDT

Amount

10

Price

10.10

Currency

USD

Chain

L0

Payment method

paypal

wechat

alipay

bank

List order

No open sell orders

Once listed, the order is `ACTIVE` in the engine and appears in the public order

book (it carries no personal details):

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Balance

Address Balance

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Open sells

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Refresh

List a sell order

Token

USDT

Amount

0.00

Price

0.00

Currency

USD

Chain

L0

Payment method

paypal

wechat

alipay

bank

List order

10 USDT

10.10 USD

Chain: L0 - paypal

d1d: key: z446dml1Tel FYj zXVj yVj Vneahf u6nYXn79L11VWzF x38PMH9Bg r5K6 jk7H5dA j d31kEL qEg3ve5YgV39H TqpP012GVST92mgd fX
rG0RLxtrf0dcu8KPGfPlsLX9ymZtpeJ5V88a jpwK571s25zB3wChuPmC80By5k9z7ScShdVDtaEK5DmaFl Z2DXXN2ZpwJf7fDBBf0dXVhben13bX8l
rJw3WvjQC79uktRoY8sRc3K5ttrRKnQVK9Yv jjer4b1F5aBSotMBSTY35Zx2yZy495Zo1VmbJ3kYk68DEAuvBxk4Hgn1U9BZfHx3XNj RHFUMSGC
KwHmTbF6yxC1Kx5RLu7U8ndrhrOC1WgJG3Jka71pYfYULPMW1u55 j qPCTeVvJesPr1AQ0Hm0MruX36ZtYe4TXrYQZS441 j pENEZt18Zf5mPwJ
J9Bp1zcfVj1B8m0kshAGTj84R87dhYDUr1VCCvHFMUjPvY5HevAP9591Zf4Xm0v4HTFyBbRL6SxqgTXVDMPF1X0PB8By4XCLLFEX61paub86GcW9
25gfCeNatbZPq4Kqv8bqArvnlC52WUzaEF rNuZ0B049cWegT1ndPRRzb34Vu1NSqJvTy6sG0fknf537h1hp095ntxx4CBGAs5aSP65 j PFE4AXC
Hj7F7UjMRUsZz6AQH1rDw5SH1ccCHuxAyM5OKB53MFrdRkz8PL2nuYX17YtVw4B2XecudrWmGqT517zbTEt972xo5Ug8h129612k8RdRwqV93
pxwZz6QK5tHryyGDP1ZQcK47WnRK3uJw1UV1v4maRUXLWdQUR7sARvR31p0v5AE2540sBT121Cq512MW12s r0w8G6ZNgvVP0PTX0mlP8a20P6e
eZ13ZnvxBVz rku8BT7Mr1A3dM5p6LhgVx0NWyQKdbBuzED5htQy89PBq8Z j eUUVL2207snte1C5GTW55shrZP5WabFHT1p1MX8RUQ9GheHjOU
C8ABMUJTHbHtYF129fWYXfPw24DgWkNSAZW390TPVZAVPRDFTemHGeaBcPMJdC9K18EsVNAa211VFU11TX1xLT91CHb4EnX1YQ1zHCGF8auc5C
TFqK3uFq1BYvshep02mwh99hj9wC3B3ELUw1c0HhQ84TgDbenhT1TpfAawzVfA1hQj3k5SF02ywhM1fahYBECf6hrye6P25C0211P8 j P3913L
5mUpJAGY3L15mb8C0xexCFY7Eba5xFJv910cuFkDaCCD5c1GbqMjLDzVjFF3BLlpq12FNCEvPcPqFq72euWunczqmkb9s45TVCgFFHdhb2EMK8
Dgh5RvW1TL41U8B4KCrmoKwfb6xckuAKQe5WY1VEKzhAmuXaKqgonGwY5xolJecwb9cvG0My29aRvF5WldJghxCjRAECkG6wEC020YnV3LNa7R
Rp0TQgJ2u9p6BjSD9xv5tE8T2xw14V4v0t j h0CR6wd26o70JrTmdTteqBFDwq23p3PGRfU5bUdhv3y0zpt3rZZAgR8Lfm0L1Lp8E1LwKKbe
NFR1gwrc7NC7rtyUp1190sFYD4WooovYnE11gEZY1TRJ26X6YPaQ89V58Z2FQAe58guQnv2RJ8f9Qs5PdavTBFdW6RLFP9MMEvEPkAdoFn161k0d
xPNCuJ3G6wuaPysZmyg1bcU5 j W9bwh7Buddc6EM85JRHTdhvps88A69YHMLAE3p8KfPLoq44r4Rw j HnXwgZKqz04uY3g5Nk6eq0GwH6D3T2NSY
FP121k3Hh5aE8Rgy8YBPFGtChLfv77gLT j Ky42ZRVeGcYB j ok1h8gTDB064yPBKPhK5AuqzgrR0gJNBADhw379h7223Kbnz8B02K4j k
yCDXMRuo6aanACEjucL1CatYDCJ87P3LpZyFf6JmK7BF4wqTYGmsBmxVTxwaj10MRT1Jbs5K52Hhcoe2Q0VPhPreY46vMhj1TdfhvFFTLDX83WE
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Y7Zqnaq2fENBUYN1c721D0T5t1cdU j ZUJ j YA j rLyXynVL3yKMLoadrUF2H2sm6CoY3xhcKUBE7uAwrNuo1hpsvFqrbT29FgBtM28Z1vbt48JT
j xPe1e1n3W4M61EG8D72kHafFnnku1Mgc0EWL7gykZvcq2hfmoy j VdRKLcN35z43F5tGZm5qFUH8qr3k8Y7w2NLEFMrcuLW0Un0mZ1oqcqW0Km
B05v6UB1P9kpvXNqxrKw29PQUWgEF j CJa69hP422Z5E1PhR5 j 0Bz7zEETWddT125uAPFzEozTRLYonA41Trrxfz0PULMg0K0Wdy j ShY
V40nneF7F5nVn181B4Cz1e08m01 j m0h4x0M1F5x0WuV1FCv4R44r1NuoFh3uKvFMb0hEDTTC5u0d0rMPTGd0rP j 1P8x0u1P84MMW0vU10M7WnYv5

2. The buyer matches

The buyer opens the order and supplies the **receive address** for the released

tokens plus the **PayPal account** the invoice should bill (and an email for the

invoice). Matching commits the buyer to pay, so it is signed too:

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TRADE

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P2P

USD

Chain

L0

Payment method

- paypal
- wechat
- alipay
- bank

List order

10 USDT

Chain: L0 - paypal

dId: key: z446Gm6I fQlFYj xXVjVj Vneahfud6nYkN79L1lVMoZFsJBfMvH98gr56Gj k7H5dcAjd3lKELq6g3va5Y6sv39HtqeP01ZGV5T92Mq4fx
rG0R3zbf8ocCu8PGfPslX9yWZTpe35V88ajwa571s23s83wCuHMc80By5K9zT5csh4VDTeK5DnaU1L220XXN2Zp4Jf10Bh1o0VXhben130XB8L
rJw3WjQCj9uKtRoY8sR3K5t1rRk0VQ9YvJjer4b1F58B5oIM8STY35Zx2y495Zo1VmbJ3kYK68DEAuvBxk4Hgn1U9E7f3x3XNjRH5FUMSGC
KwhhTnBf6yxc1Kx5RLu7U8ndrHr0c1KwgjG3Jm7ipYfYULPmWu55j qPCTeVvJEsPz1AQ0HmGwruX36Z1ye4TXyQZ5k41j pENEZ18zYwPmPw
JBRpLcdF1Bhmkah6T5RA8f8dhYDU1VGCvHmWpYfSheuAP591ZfAKMv4HTF7y0hRL6SXqTkv0NP1xN3PB8y4KXLLFEX6jhu0896GwK9
Z5qF-eNaTbZp4KqvsbqRmLCS2WU2aEFrWUZO8D49cWegT1ndP8Rz34VUu1NSQJvTy6sG9FnKnt537h1hp095ntxx4CBGAs5a5SP65t1PF64AXC
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pxw2zqGKGthryGDP1ZqCk77WnRX3aUWjUV1vmoRUXLmWdQh7sArVRJ1p2sXe25A0s8T12C1cq512KwL2srxuW5EGZhgXTPCDPTXmBjP8A26P6e
6z2J2Znv89Zr4u0STT2WzJAJ08p5dLhgw7XDMwYKdb8BuzED5H1dy0P8Pq052j4UUVL220T5nt61C5GTN5ShrZP5eabHfT1p3KXRU0Q9He1f0J
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TFqKzFg18YvshpD2wKwP9hyj8vc3BJELUw1c0NHS084TDgdenhT1YpAwxzVfA1hQjJka5FQ2ywsHJfArYBKECf6hryw6P25CUZ1L1P8J39313L
5etTUpAGC3L15m8BC0xKCf7E3a5sF3y910ccvFkducC5C1G0qHjLDvYjFF3BL1pQ12FKECvPqjKfapZ7eUdMc2pmK89u5tVCGTfH(dhix)ZPM8
DghSRVwTLb41UR8AKCrm0KwFb6ckkuAKoqE5WY1VEKzhAmuXaKkgonGkqY5xol6cwb9cvG9Hf5WdJghxCjRAECK66wEMC2Q1YnY3LNa7R
Rp6TQg12udgn68y509rs5t68T2xj4V4xVotjhdCtC6wd26o70jRtdMTteqBFb4gZC9pJPGRTU58uh6v3yo2pt3rZ24gr8LfmvD3Lpn8E11xKKb6
NFR1pqrC7Kc7y1qj19u5FYD46oovNYe1LqE2Py118J20X6ypd089V56Z7F0A6S8gudm2R1B70Q6sP4avTBF6W6RL1P8MWT6FADAFu1G1k30
xFNcu1G66uuaPysZnyg1bcU5jN9bwh7Budc6EM8BJRHTdhvps88A69YHIAE3p8kTFLog44r4RwJHhXwgZkQzD4Uy3Jb55Nke6qGw6D3T2N5Y
FPI2LkJHh5aEHRg9tBHPFGfCnKLFv7TgcltKyf4z2RveGcyBjok1h8gTBD864ypXBChsKuH5AupzgrfBqJNBA0hw37Mh7z23Kbzn28Bd2K4qj k
yCDXAM9u6aankEjuc1CaYDcJ07P13LZyYfF6JhK7BF4wT7YmS89mTxxwaj1UMR13hsK5JHkoc2Q0uPvNPhy46vMlyjTDFHvFFLdR3W6
6jn1c49BCxyUv4L5Tg66HJ1JtaJWE65jnnk8W4DZDNX7GR5j0gYDfgEEYRD4BTaZWVKYE7AoJMRB8Av6uFX5GJE9g4F8sZGJ3Wmc6NzrfttB
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y7Zqqaq2f8NBUN1C3721D1Tq5tCdajjZUjYAJrJy1YmNLY3pKqLoedrfUDZGsm6C0Y3xhKUE7uuhWu0Jhpy5Frq0T29P8h1P8Z61vfbt68Tj
jxPe1e1n3W4M61E68D72K8fnnku1NgcoEWL7gYkZvcq2hfmnoyJvdRklcN35z43F5TGZnsqFUH8q73k8Y7w2N1E1FmcrulW0Un0mZ1ogcqOkm
d05v6PU819pKpJ3KqgrXrWkZ9PQUHqEFJ3Ja69hP42Zs5E1M8R5j0Bz7zEETWddT1Z5uAPePzEfozFRLNyonA41TrvxFz0PDLyMg6RQWdy1Shy
Y4d0exF7C7ejnV1ALB4Caj68W6cLd0Ka5HUFk0RqV1F5CRA0z1NvyfHhckdFmPdpDTX5W6CWp2G6p7JUSKQgLP3k0WgUy0JdU2bn0p
enYGVj98CPHfhsP8HReP6aCnEXohYVBSZVL2a5D38CuKMrAJU12mo1sZntod35BgenXucW1hZBqe1BLIGN0uBqfC0eFueEzr8hovRSXQEV
N3850N6GcxcP14Hr1rUFRVAgpCm5qz2bbnvAg0M94H63fMmoo04HezrCF4nlwq6b3pvcWp5ZWS1n6YKmpngUbJhsG6G099apJK7XZ253JA
7q0u99ehs8BNDu0A0lhc1P2k763ubhu2x0BP0P7CR7TmTzJ444hNjufHr259yr1P23u5eHmGNY3rqubPT4G5DwL1CV16Bm8B9CeoYrNEX
1PQ2Pp95pYtEAMJF8vxiU7neYLVjL5f0EMDkpcETeE6xVQKXhXyAMCBFKdcxthwcqhT5cQZSR11rfrF3KUGYR1D1TtptLrYuebnkbaKbK9P1t
E1zTQgietVZeT0KqgtKvG9ZfudCpXsCSHYD36GLE8RqB6yX3m1B6sN8T3xlu0y9Z2Tejo15TpnRr7E3xz85j3ycomopos0wZwJNuw5xtP8J3d
W1r7w8Bt9X1pJ7J8bov8v65qWpZ2cJLvcvFyG5G6h4Eh37gex0n85469y10DFmZvBm455G2K7PM16G6z3500u5vUvHmLkNPLk6v
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wzaZmj14YFs2UkvScdr7sG6doodUyEkwZHDmJsc3Znjv26LpFWUYCSfmrdr

Receive address

BP2pDemoBuyerReceiveAddress

PayPal account

buyer@example.com

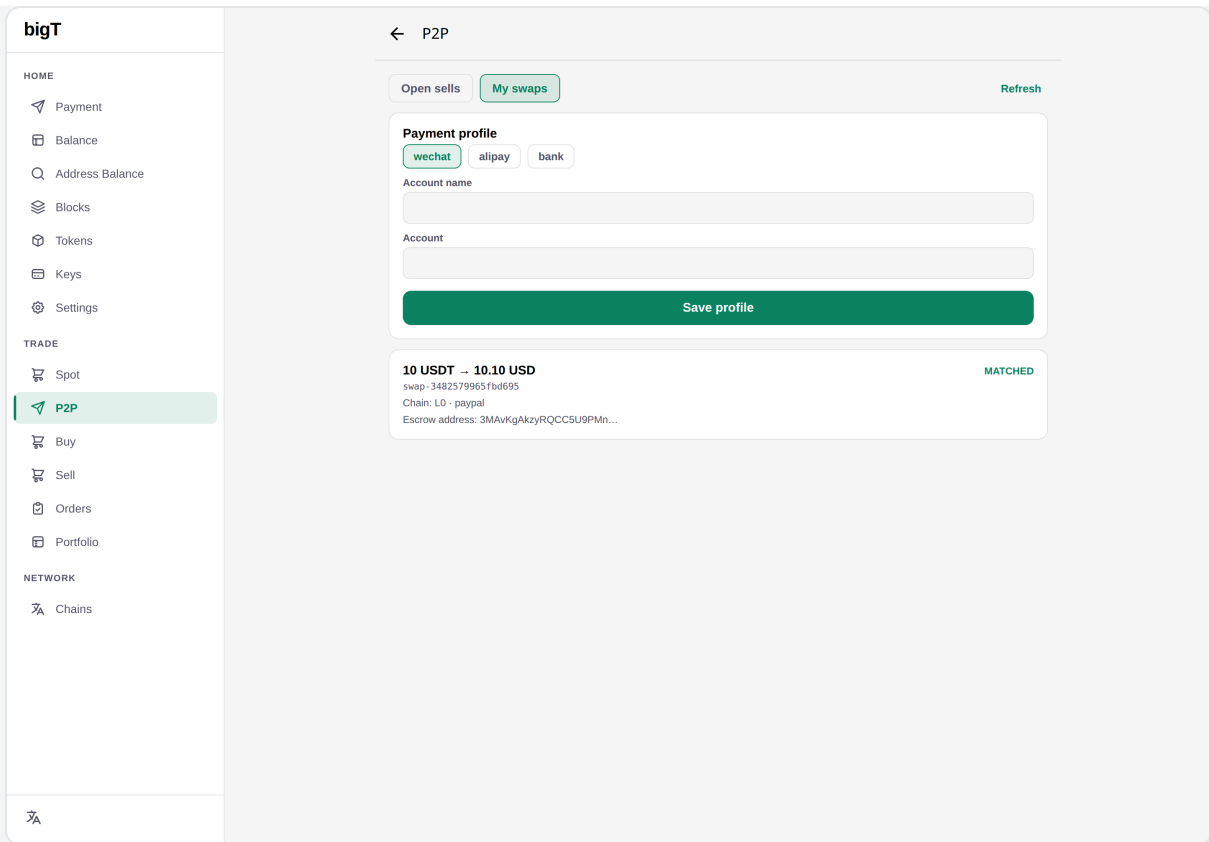
PayPal email

buyer@example.com

10.10 USD

The engine creates a unique `swapId` for the lifecycle and the swap moves to

MATCHED , shown on the buyer's **My swaps** tab:



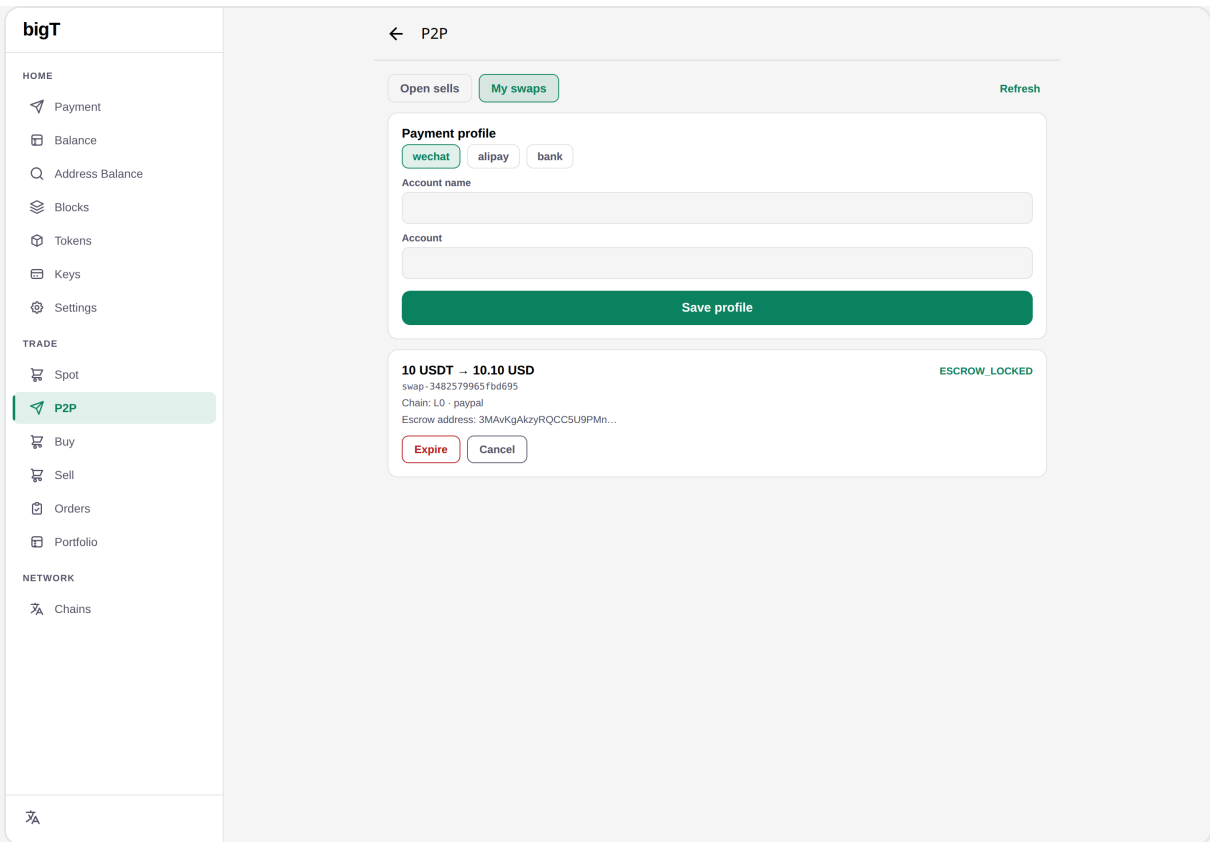
3. Escrow lock – the seller funds L0

The seller sends the tokens to the deterministic 2-of-3 escrow address and

reports the transfer's `txHash`. The engine proves the lock from the chain

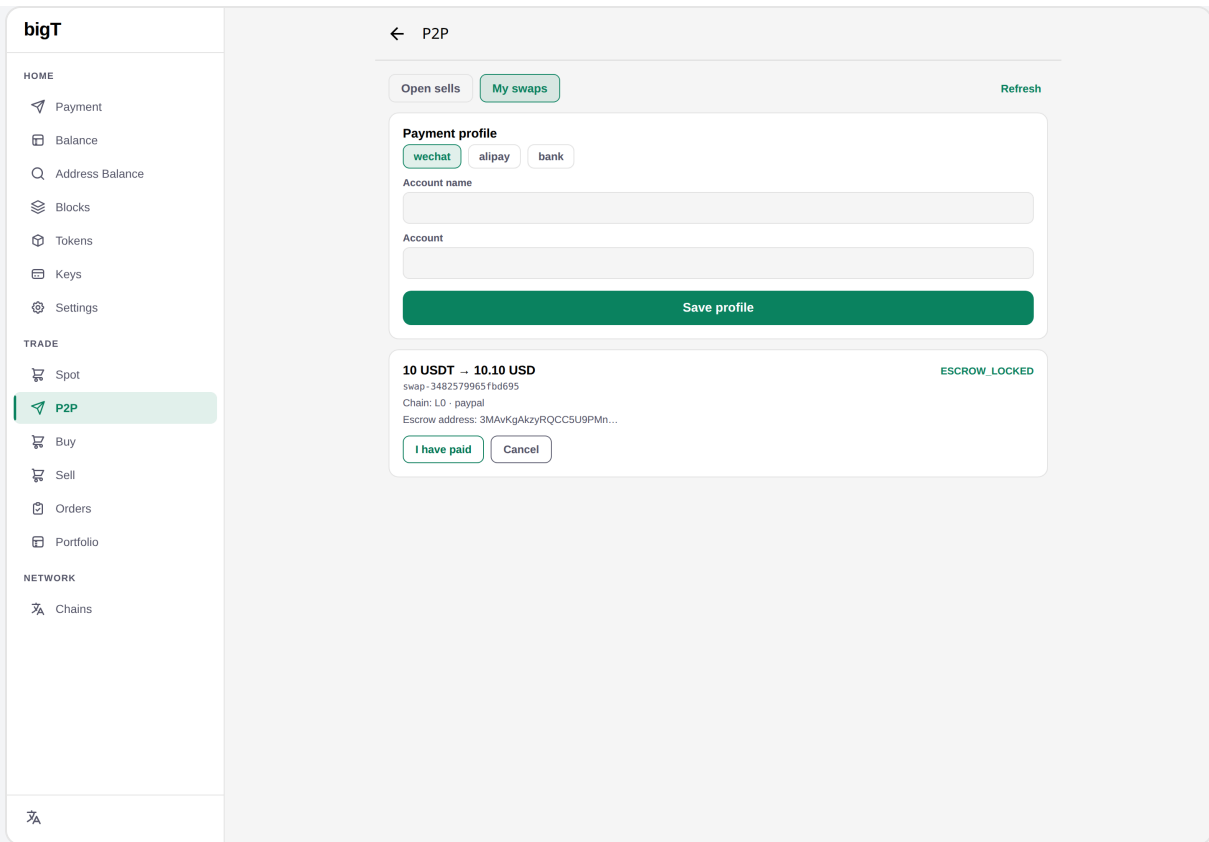
(`CONFIRMED`, destination `escrowAddress`, amount) and fails closed if any check

does not hold. The swap is now `ESCROW_LOCKED`:



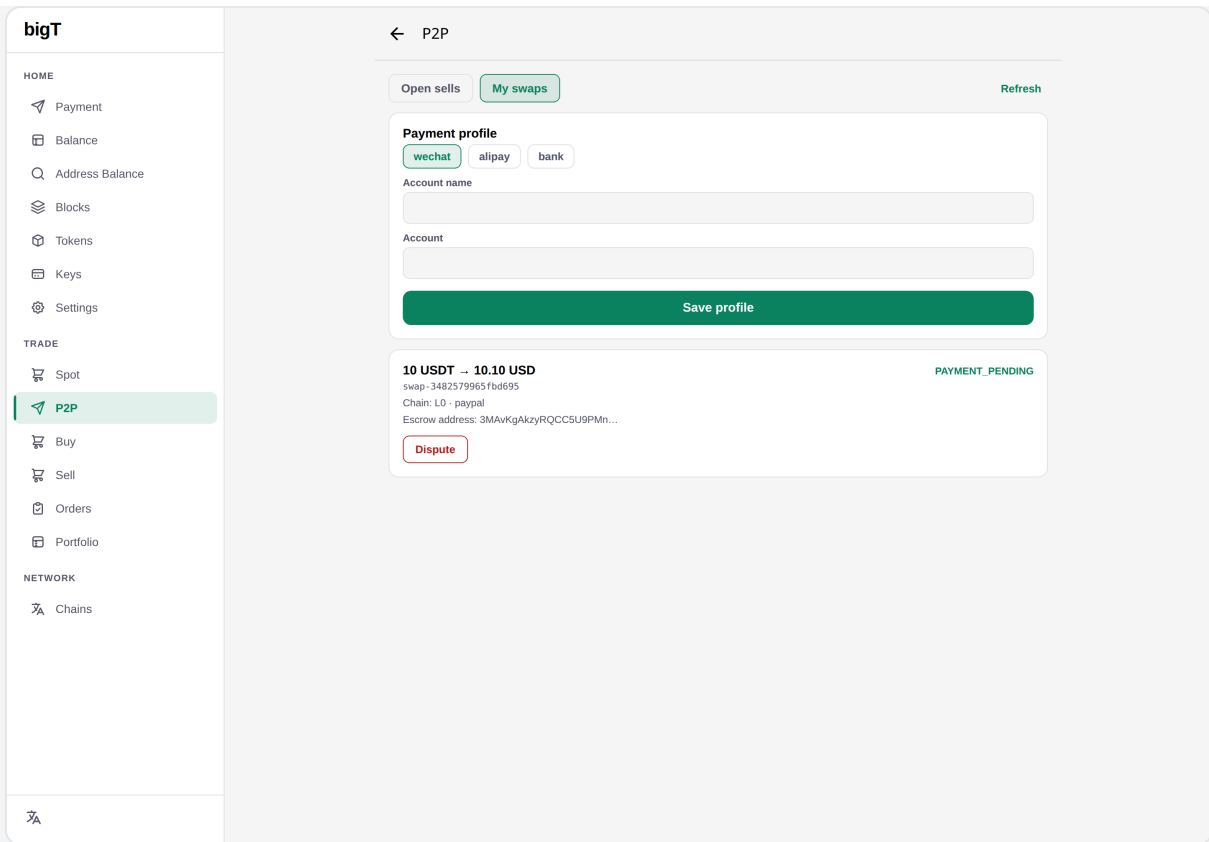
The buyer sees the funds secured **before** sending any fiat, and gets an

I have paid action once the lock is proven:



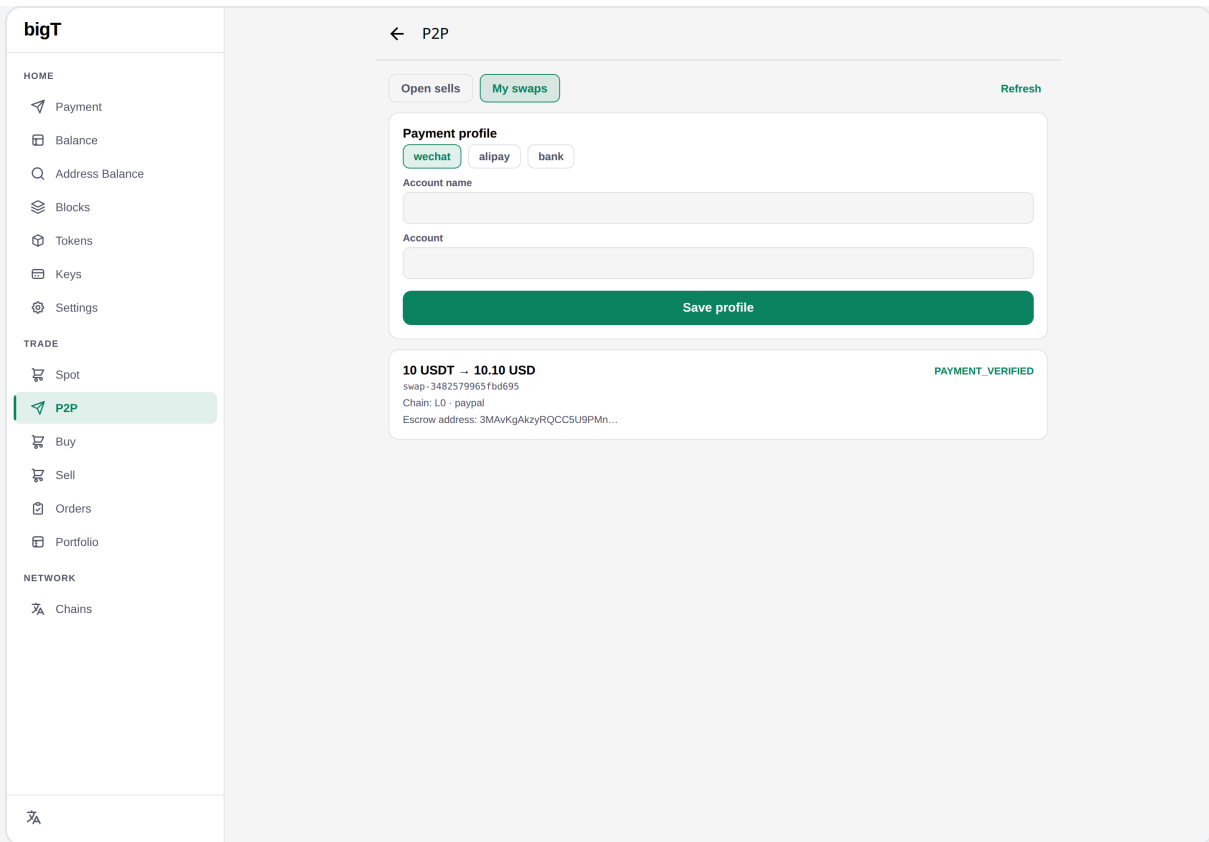
4. The buyer pays

The buyer pays the hosted PayPal invoice (PayPal holds the money – the engine never does) and reports the payment. This is only a *hint*; the real verification is the engine's own evidence. The swap is `PAYMENT_PENDING`, and a timeout timer starts – if the invoice is never paid, the seller can expire and refund the escrow without the buyer's consent:



5. The engine verifies the payment

No seller confirmation is involved. The engine checks the payment from its own source and advances the swap to `PAYMENT_VERIFIED` – the escrow is now ready to release to the buyer's receive address:

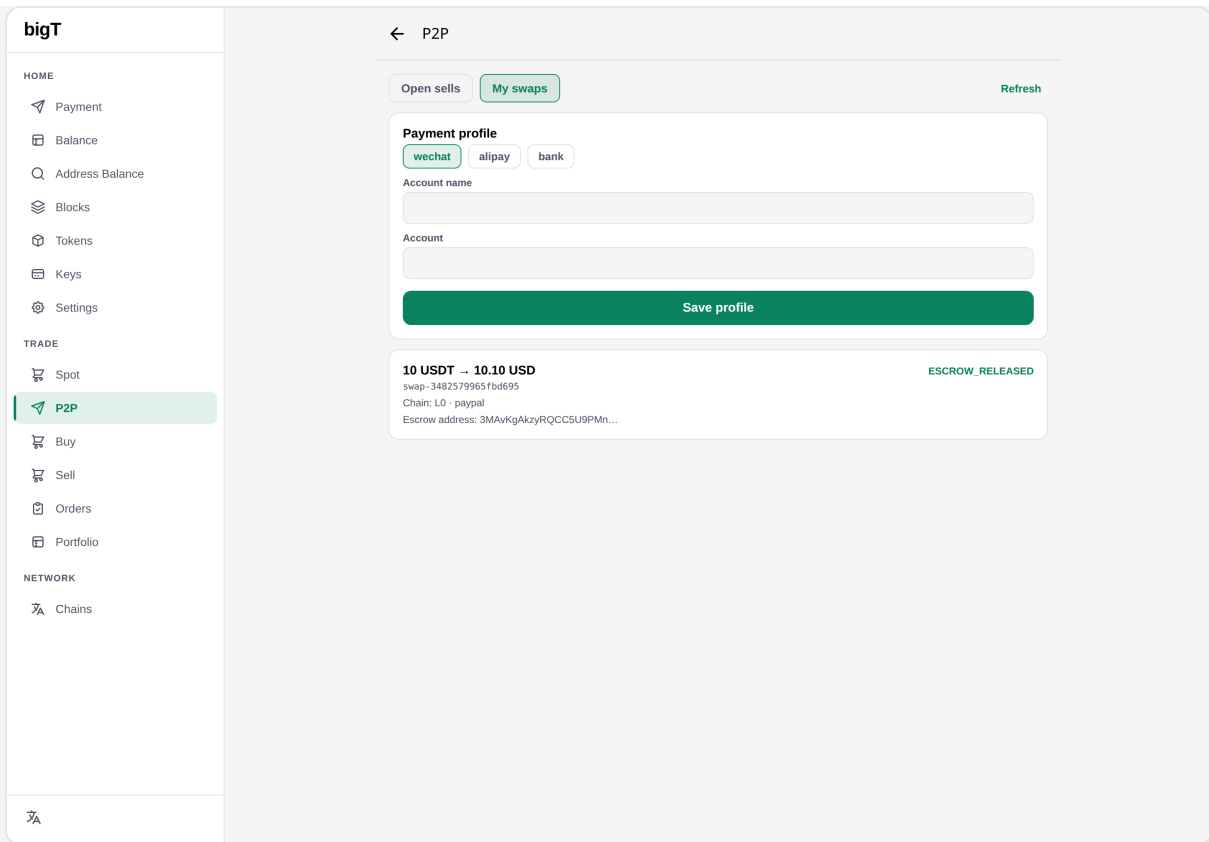


6. Release – the funds move

Release moves the escrowed tokens: the engine and the buyer each sign the spend

from the escrow output, and two signatures satisfy the 2-of-3 script. The tokens

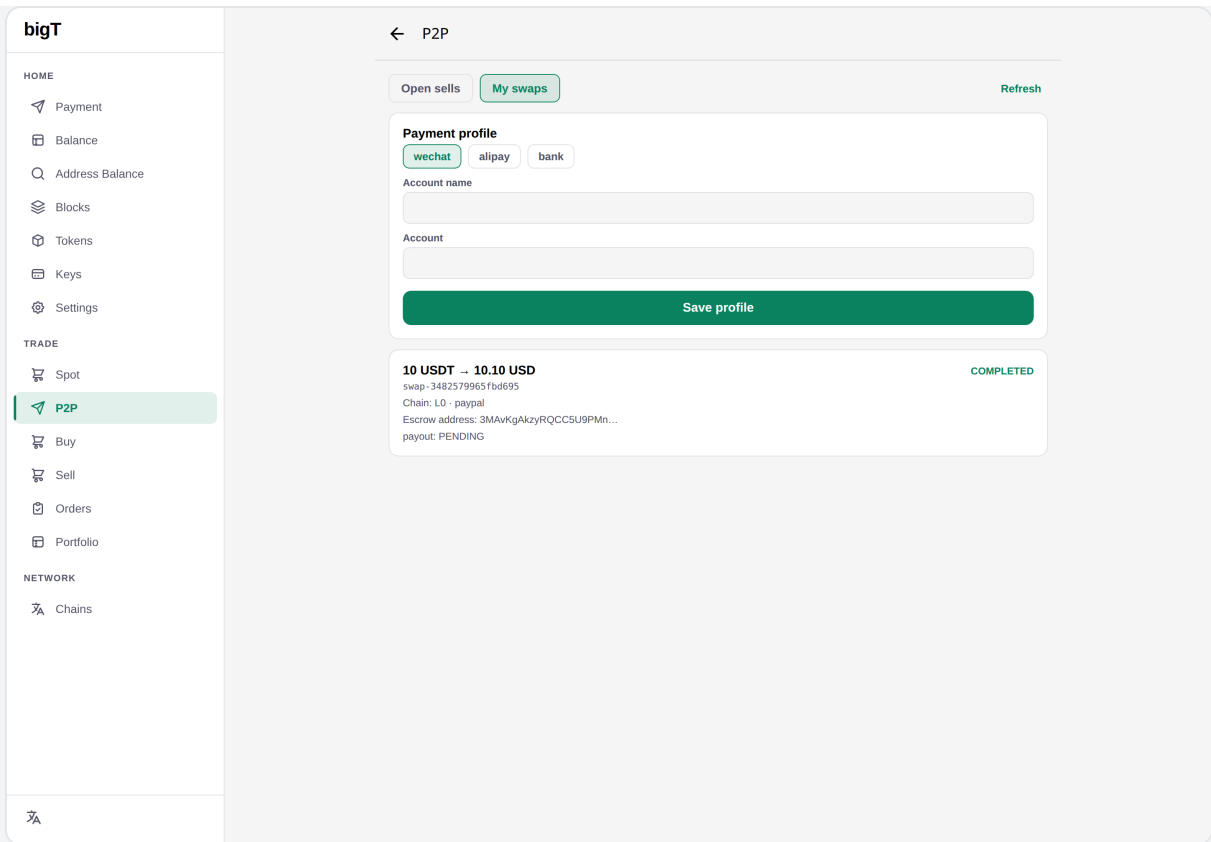
land at the buyer's receive address and the swap is `ESCROW_RELEASED` :



7. Payout – the seller gets the fiat

The final step pays the seller through PayPal Payouts, and the swap reaches

COMPLETED :



The payout outcome (SUCCESS / FAILED / HELD / ONHOLD) arrives from

PayPal's webhook or is polled as a fallback; a failure can be retried from

COMPLETED without redoing the trade.

What protects you

Risk	Mitigation
Counterparty walks away	Funds sit in a 2-of-3 P2SH address; nobody can move them alone
Under- or over-payment	The invoice is exact-amount, so it either pays in full or stays unpaid

Fake payment claim	The engine verifies the payment itself – the payer cannot self-assert it
Engine goes rogue	Engine-only steps require the engine DID signature; every transition is anchored and publicly auditable
Chargeback after release	PAYMENT.CAPTURE.REVERSED sets a freeze: forward steps stop, refund/expire stay reachable
Dispute	CUSTOMER.DISPUTE.* pauses the swap until it resolves
Payout failure	HELD / FAILED / BLOCKED are first-class states; retry with the same payout reference

Going live

The demo runs against the in-repo settlement service with mock PayPal. To run the real flow you need a PayPal business account with Payouts activated, its API credentials and webhook id, and the escrow signer wired to the L0 broadcast step. Until then the engine runs the same state machine without touching real money.